# LOOP CHECK SHEETS - DEMO-GPF Gas Processin

Document GPF-DEMO-LCS-001  Rev A-draft

**DRAFT - generated from the engineering database. Not c**

| Loop | Tags |
|---|---|
| 100-AT-017 | 100-AT-017 |
| 100-ESDV-001 | 100-ESDV-001, 100-ZSC-001 |
| 100-FT-010 | 100-FT-010 |
| 100-LT-008 | 100-LT-008, 100-LV-008 |
| 100-LT-013 | 100-LT-013 |
| 100-PDT-015 | 100-PDT-015 |
| 100-PT-001 | 100-PT-001 |
| 100-PT-003 | 100-PT-003 |
| 100-PT-005 | 100-PT-005 |
| 100-PT-011A | 100-PT-011A |
| 100-PT-011B | 100-PT-011B |
| 100-PT-011C | 100-PT-011C |
| 100-PT020 | 100-PT020 |
| 100-PV-004 | 100-PV-004 |
| 100-TE-018 | 100-TE-018 |
| 100-TT-007 | 100-TT-007 |
| 100-TT-016 | 100-TT-016 |
| 100-XV-012A | 100-XV-012A, 100-ZSC-012A |
| 100-XV-012B | 100-XV-012B, 100-ZSC-012B |
| 200-ESDV-001 | 200-ESDV-001, 200-ZSC-001 |
| 200-FT-018 | 200-FT-018 |
| 200-FT-025 | 200-FT-025, 200-FV-025 |
| 200-LT-012 | 200-LT-012, 200-LV-012 |
| 200-LT-013 | 200-LT-013, 200-LV-013 |
| 200-LT-024 | 200-LT-024, 200-LV-024 |
| 200-LT-031 | 200-LT-031 |
| 200-LT-041 | 200-LT-041 |
| 200-PDT-026 | 200-PDT-026 |
| 200-PT-011 | 200-PT-011, 200-PV-011 |
| 200-PT-021 | 200-PT-021, 200-PV-021 |
| 200-PT-022 | 200-PT-022 |
| 200-PT-031A | 200-PT-031A |
| 200-PT-031B | 200-PT-031B |
| 200-SDV-042 | 200-SDV-042, 200-ZSC-042 |
| 200-TT-015 | 200-TT-015 |
| 200-TT-016 | 200-TT-016 |
| 200-TT-017 | 200-TT-017 |
| 200-TT-023 | 200-TT-023, 200-TV-023 |
| 200-TT-027 | 200-TT-027 |
| 300-ESDV-041 | 300-ESDV-041, 300-ZSC-041 |
| 300-ESDV-042 | 300-ESDV-042, 300-ZSC-042 |

## LOOP CHECK SHEET - 300-FT-003

DEMO-GPF Gas Processing Facility (fictional) | GPF-DEMO-LCS-001 Rev A-draft

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| | |
|---|---|
| **Loop** | 300-FT-003 |
| **Service** | Compressor suction flow (anti-surge) |
| **Unit** | 300 |
| **System** | BPCS |
| **P&ID** | GPF-300-PID-001 |
| **SIF** | - |

**Instrument data (from instrument index and I/O list)**

| Tag | Service (type) | Signal \| Range \| Ex | I/O address | Card |
|---|---|---|---|---|
| 300-FT-003 | Compressor suction flow (anti-surge) (FT) | 4-20mA HART \| 0–400 kSm3/h \| Ex ia IIC T4 | CAB-BPCS-03/R1/S01/CH03 | AI-HART-IS IS |
| 300-FV-003 | Anti-surge recycle valve (FV) | 4-20mA HART \| 0–100 % \| Ex ia IIC T4 | CAB-BPCS-03/R1/S05/CH01 | AO-HART-IS IS |

**Test steps**

| No. | Tag | Action | Expected result | As found | As left | Pass / Fail | Initials |
|---|---|---|---|---|---|---|---|
| 01 | 300-FT-003 | Confirm field device 300-FT-003 is installed, tagged and terminated per loop drawing; check continuity and insulation to CAB-BPCS-03/R1/S01/CH03 (AI-HART-IS, IS). | Correct device, tag plate, terminations and cable; no earth fault. | | | | |
| 02 | 300-FT-003 | Inject 4 mA (0 % span) at the field terminals. | Display reads 0 kSm3/h (± 2 kSm3/h). | | | | |
| 03 | 300-FT-003 | Inject 8 mA (25 % span) at the field terminals. | Display reads 100 kSm3/h (± 2 kSm3/h). | | | | |
| 04 | 300-FT-003 | Inject 12 mA (50 % span) at the field terminals. | Display reads 200 kSm3/h (± 2 kSm3/h). | | | | |
| 05 | 300-FT-003 | Inject 16 mA (75 % span) at the field terminals. | Display reads 300 kSm3/h (± 2 kSm3/h). | | | | |
| 06 | 300-FT-003 | Inject 20 mA (100 % span) at the field terminals. | Display reads 400 kSm3/h (± 2 kSm3/h). | | | | |
| 07 | 300-FT-003 | Connect HART communicator; read tag, range and PV. | HART tag 300-FT-003, range 0–400 kSm3/h; PV agrees with display. | | | | |
| 08 | 300-FT-003 | Simulate input below 3.6 mA (open circuit). | Bad-quality / instrument fault alarm raised; value flagged invalid (NAMUR NE 43). | | | | |
| 09 | 300-FT-003 | Ramp input slowly (decreasing) from the normal value to 8.8 mA (120 kSm3/h). | L alarm, priority High | | | | |
| 10 | 300-FV-003 | Confirm field device 300-FV-003 is installed, tagged and terminated per loop drawing; check continuity and insulation to CAB-BPCS-03/R1/S05/CH01 (AO-HART-IS, IS). | Correct device, tag plate, terminations and cable; no earth fault. | | | | |
| 11 | 300-FV-003 | Command output to 0 % from the operator station. | Loop current 4 mA; valve travel 0 % (observe at valve). | | | | |

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| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 12 | 300-FV-003 | Command output to 25 % from the operator station. | Loop current 8 mA; valve travel 25 % (observe at valve). | | | | |
| 13 | 300-FV-003 | Command output to 50 % from the operator station. | Loop current 12 mA; valve travel 50 % (observe at valve). | | | | |
| 14 | 300-FV-003 | Command output to 75 % from the operator station. | Loop current 16 mA; valve travel 75 % (observe at valve). | | | | |
| 15 | 300-FV-003 | Command output to 100 % from the operator station. | Loop current 20 mA; valve travel 100 % (observe at valve). | | | | |
| 16 | 300-FV-003 | Remove output signal (loop disconnected). | Valve moves to its fail-safe position as stated on the valve datasheet. | | | | |

**Test equipment**

| | |
|---|---|
| Calibrator / ID | |
| Calibration due | |
| HART communicator / ID | |
| Insulation tester / ID | |

**Remarks**

**Signatures**

| Role | Name | Signature | Date |
|---|---|---|---|
| Performed by (Contractor) | | | |
| Witnessed by (Company) | | | |